A One-Pixel Counterexample to Automatic Aubin Regularity

Candidate full counterexample for arXiv:2202.04680

11 August 2026

Status. Candidate full counterexample, likely valid, pending human review. The Aubin condition in the source's convergence question is not automatic for the proposed functional class.

1 Source and exact question

N. Gruber, J. Schwab, S. Court, E. R. Gizewski, and M. Haltmeier, *Lifting-based variational multiclass segmentation algorithm: design, convergence analysis, and implementation with applications in medical imaging*, arXiv:2202.04680v3 [1]. On source PDF page 17, after deriving its nonlinear primal–dual algorithm, the paper asks whether the Aubin property and a dual-variable smallness condition from the convergence analysis in [2] hold for its functional.

> Using the ingredients computed above, we obtain the following proposed Algorithm 1 for generating a sequence $(\boldsymbol{u}^n, \bar{\boldsymbol{u}}^n, \boldsymbol{v}^n, \boldsymbol{w}^n)_{n\in\mathbb{N}} \in (\mathbb{H}^K \times \mathbb{H}^K \times \mathbb{H}^{2K} \times \mathbb{H}^{2K})^{\mathbb{N}}$, where $\boldsymbol{u}^n$ approximates the segmentation mask and $\bar{\boldsymbol{u}}^n, \boldsymbol{v}^n, \boldsymbol{w}^n$ are auxiliary quantities derived from the PDHG algorithm. The application of the convergence analysis [15] requires the Aubin property as well as a smallness condition for the dual variable. It is an open problem whether these properties are satisfied for our functional.

Figure 1: The exact open statement in the source, PDF page 17.

The answer is negative if the question is read as asking whether these conditions hold automatically over the proposed discrete model class. A degenerate but fully admissible one-pixel instance already violates the Aubin property, even with zero dual variable.

2 The saddle map used by the convergence theorem

Write the source's finite-dimensional objective in the form

$$\min_u F(\mathcal{K}(u)) + G(u), \qquad \mathcal{K} = (\nabla, \mathcal{M}), \quad G = \delta_{\mathcal{U}},$$

where $\mathcal{U}$ is the pointwise multiclass simplex and

$$F(v, w) = \lambda \|v\|_{2,1} + \|w\|_{1,1}.$$

Thus $F^* = \delta_{\mathcal{C}}$, where $\mathcal{C}$ is the product of the corresponding closed dual balls.

For a reference primal point $\bar{u}$, the convergence analysis [2] uses the set-valued linearized saddle map

$$H_{\bar{u}}(u, y) = \begin{pmatrix} \partial G(u) + \mathcal{K}'(\bar{u})^* y \\ \partial F^*(y) - \mathcal{K}'(\bar{u}) u - c_{\bar{u}} \end{pmatrix}, \qquad c_{\bar{u}} = \mathcal{K}'(\bar{u})\bar{u} - \mathcal{K}(\bar{u}).$$

Its inverse is required to have the Aubin property at zero for the reference saddle point.

3 Idea of the proof

Choose a one-pixel, two-class instance with both feature values zero. Then both the discrete gradient and every nonlinear discrepancy term vanish identically, so the linearized saddle map is exactly the product of the normal-cone maps of the primal simplex and the dual ball.

At the simplex midpoint, zero is a normal. But an arbitrarily small normal perturbation in the tangent direction $t(1,-1)$ selects the distant vertex $(1,0)$ as its unique inverse image. The input perturbation tends to zero while the inverse-image distance stays $1/\sqrt{2}$, contradicting the metric-regularity estimate equivalent to the Aubin property. The reference dual variable is zero, so this failure persists even when the separate dual-smallness condition is satisfied maximally.

4 Counterexample

Theorem 1. *There are admissible dimensions, feature maps, and a saddle point for the source functional such that the reference dual variable is zero but $H_{\bar{u}}^{-1}$ does not have the Aubin property at zero. Consequently, the two convergence conditions named in the source do not hold automatically.*

Proof. Fix any $\lambda > 0$. Choose a one-pixel image, so $N_1 = N_2 = 1$, and two classes. The primal variable is $u = (u_1, u_2) \in \mathbb{R}^2$, constrained to

$$\mathcal{U} = \{u \in \mathbb{R}^2 : u_1, u_2 \geq 0,\ u_1 + u_2 = 1\}.$$

Choose both feature values to be zero. The one-pixel discrete gradient is identically zero. Moreover, the source's smoothed class average is zero:

$$A(u_k) = \frac{\langle u_k, 0\rangle}{|u_k|_{1,\varepsilon}} = 0.$$

Every component of its nonlinear discrepancy map has the form

$$m(u_k) = u_k\big(A(u_k)\mathbf{1} - 0\big)^2 = 0,$$

and the same holds with u_k replaced by $1 - u_k$. Hence

$$\mathcal{K} \equiv 0, \qquad \mathcal{K}' \equiv 0, \qquad c_{\bar{u}} = 0.$$

The saddle map therefore reduces exactly to

$$H_{\bar{u}}(u, y) = \big(N_{\mathcal{U}}(u), N_{\mathcal{C}}(y)\big), \tag{1}$$

where N denotes the convex normal cone.

Let

$$\widehat{u} = (1/2, 1/2), \qquad \widehat{y} = 0, \qquad \widehat{z} = (\widehat{u}, \widehat{y}).$$

Because $0 \in N_{\mathcal{U}}(\widehat{u})$ and $0 \in N_{\mathcal{C}}(0)$, equation (1) gives $0 \in H_{\bar{u}}(\widehat{z})$. More precisely,

$$N_{\mathcal{U}}(\widehat{u}) = \operatorname{span}\{(1,1)\}, \qquad N_{\mathcal{C}}(0) = \{0\},$$

because $\widehat{u}$ is in the relative interior of the simplex and zero is in the interior of every dual ball.

For $t > 0$, put

$$p_t = t(1,-1), \qquad q_t = (p_t, 0).$$

A point of $\mathcal{U}$ has p_t in its normal cone exactly when it maximizes $\langle p_t, u\rangle = t(u_1 - u_2)$ over the simplex. The unique maximizer is $e_1 = (1,0)$. Also $(N_{\mathcal{C}})^{-1}(0) = \mathcal{C}$. It follows that

$$H_{\bar{u}}^{-1}(q_t) = \{e_1\} \times \mathcal{C}, \qquad \operatorname{dist}\big(\widehat{z}, H_{\bar{u}}^{-1}(q_t)\big) = \|\widehat{u} - e_1\| = \frac{1}{\sqrt{2}}. \tag{2}$$

Suppose, toward a contradiction, that $H_{\bar{u}}^{-1}$ had the Aubin property at 0 for $\widehat{z}$. The residual metric-regularity estimate used in [2], applied with $u = \widehat{z}$ and $w = q_t$, would give for all sufficiently small $t > 0$

$$\operatorname{dist}\big(\widehat{z}, H_{\bar{u}}^{-1}(q_t)\big) \leq L \operatorname{dist}\big(q_t, H_{\bar{u}}(\widehat{z})\big) = L \operatorname{dist}\big(p_t, \operatorname{span}\{(1,1)\}\big) = L\sqrt{2}\,t$$

for some finite L. This contradicts (2) as $t \downarrow 0$. Thus the Aubin property fails.

Finally, $\widehat{y} = 0$, so every projection of the reference dual variable onto the nonlinear dual block is zero. The informal dual-smallness requirement is therefore satisfied as strongly as possible, but the Aubin requirement still fails. □

5 A second structural obstruction

The cited convergence theorem also assumes strong convexity of F^* on the nonlinear dual subspace. For the source functional, $F^* = \delta_{\mathcal{C}}$ on that block. Pick two distinct interior points of the relevant dual ball. The indicator is zero at both points and their midpoint, so no positive strong-convexity modulus can satisfy the defining midpoint inequality. Thus this hypothesis is not automatic either.

This observation is independent of Theorem 1. The counterexample itself directly answers the source's named Aubin question.

6 Verification and scope

1. The construction uses only dimensions and feature maps allowed by the source; the smoothing parameter keeps every denominator positive.
2. The affine residual $c_{\bar{u}}$ vanishes because the complete operator, not merely its derivative at one point, is zero.
3. The fixed distance in (2) is exact in the product Hilbert norm because $0 \in \mathcal{C}$.
4. The proof uses the exact residual metric-regularity formulation from the cited analysis, including the distance to the full set $H_{\bar{u}}(\widehat{z})$.
5. The result does not assert divergence of the implemented algorithm. It shows that the cited convergence assumptions require additional nondegeneracy hypotheses and cannot be inferred from the functional's form.

7 Bounded novelty check

The local run indexes were searched for arXiv:2202.04680, the exact title, the exact open sentence, *Aubin property*, and *dual variable*. A bounded external search on 11 August 2026 used the same phrases and citation-oriented variants. It found the source, its 2024 journal version, the cited nonlinear PDHG paper, and general work on Aubin properties, but no explicit answer to this source question. The search was not an exhaustive citation-graph or MathSciNet review, so originality remains a candidate claim.

References

[1] N. Gruber, J. Schwab, S. Court, E. R. Gizewski, and M. Haltmeier, Lifting-based variational multiclass segmentation algorithm: design, convergence analysis, and implementation with applications in medical imaging, *J. Comput. Appl. Math.* 439 (2024), 115583; arXiv:2202.04680, doi:10.1016/j.cam.2023.115583.

[2] T. Valkonen, A primal–dual hybrid gradient method for non-linear operators with applications to MRI, *Inverse Problems* 30 (2014), 055012; arXiv:1309.5032, doi:10.1088/0266-5611/30/5/055012.